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Mechanical Motion Capture Suit for Robotics

The invention relates to the field of robotics and mechatronics, specifically to wearable exoskeleton devices. The objective of the present invention is to create a kinematically identical device designed both for real-time control of humanoid robots and for collecting human motion kinematic parameters for their training.

Modern robot control devices can be implemented in the following ways: via software interfaces with pre-written code, through stationary teleoperation devices, using cameras that capture the operator's pose, as well as by employing other sensor-based or hybrid methods. As a result, existing methods prove to be either insufficiently flexible, overly bulky, or limited in accuracy, which hinders the collection of high-quality data and effective control of humanoid robots. Therefore, there is a need to develop a device that combines kinematic identity, proprioceptive accuracy, and mobility, while also providing ease of use and highly precise reproduction of control movements.

A known device is described in the article "HOMIE: Humanoid Loco-Manipulation with an Isomorphic Exoskeleton Cockpit" [arXiv:2502.13013]. The essence of the device is HOMIE — an affordable exoskeleton interface with 7 degrees of freedom for the arms, a sensor glove, and a foot pedal, integrated with a trained RL policy for loco-manipulation. This policy is capable of walking, squatting, and performing arm tasks in arbitrary upper-body postures without relying on predefined movements. As a result, the device provides high-precision full-body control and enables the collection of demonstration data for training.

The disadvantages of the known device are as follows:

- lack of a structural design that allows the user to wear the device during operation;
- absence of an additional degree of freedom in the exoskeleton to measure the user's torso rotation, which is due to structural limitations and the configuration of the elements in the back area;

– absence of an additional degree of freedom in the exoskeleton to measure the torso inclination of the user in the sagittal and frontal planes, which is also due to structural limitations and the configuration of the elements in the back area;

– the degree of freedom provided for the pronation and supination angle of the hand relative to the forearm does not ensure the full range of angular rotation.

A known device is described in the article "NuExo: A Wearable Exoskeleton Covering the Full Range of Upper Limb Motion for Outdoor Data Collection and Humanoid Robot Teleoperation" [arXiv:2503.10554v1]. The essence of the device is a wearable exoskeleton that combines intuitive teleoperation and multimodal data collection. Thanks to a novel shoulder mechanism design with synchronized joints and a belt-driven transmission, the device is capable of accurately reproducing complex shoulder motions and provides 100% coverage of the natural range of motion of the upper limb. With a weight of only 5.2 kg, NuExo supports backpack-style wearing and is comfortable for daily outdoor use. Furthermore, an integrated intuitive teleoperation control system and a comprehensive data collection system have been developed, incorporating multimodal sensor equipment for various humanoid platforms.

The disadvantages of the known device are as follows:

– IMU sensors are used to track wrist rotation during pronation and supination, which introduces additional errors compared to using encoders;

– the exoskeleton lacks an additional degree of freedom to provide torso rotation of the user, due to structural limitations and the configuration of elements in the back area;

– the exoskeleton lacks an additional degree of freedom to measure torso inclination of the user in the sagittal and frontal planes, which is also due to structural limitations and the configuration of elements in the back area.

The closest technical solution to the claimed invention, selected by the applicant as a prototype, is the invention described in patent CN 119610066A "Wearable Exoskeleton Device, Control Method, and Robot Teleoperation System." The essence of this device is a wearable exoskeleton comprising a back frame with shoulder, elbow, forearm, and/or wrist modules. Each module is equipped with a sensor (encoder) for collecting kinematic data of the user's movements. The shoulder module consists of upper and lower joints connected by a curved link designed according to the shoulder's anatomy. It is movably connected to the back frame, with the joints equipped with encoders to capture the shoulder angle. The shoulder module includes a rotating shaft mounted on the link and a damper to reduce friction. The elbow module contains a joint and a frame with a fastening part, also equipped with an encoder. The forearm module consists of a link with a sensor; this link includes a rotating part and a telescopic part, the latter being adjustable according to the forearm length. The wrist module comprises front and rear brackets, a joint, a protective cover, a gear mechanism, and an encoder; the sensor captures the rotation of the wrist and/or hand. The back frame includes a structure frame, a control unit with a power module, and encoders for each joint. The controller processes the obtained angular data. The operating method involves the user wearing the device, where the sensors collect the kinematic data of the arms; this data is stored and/or used for training other robots. The robot teleoperation system includes the wearable exoskeleton device and a target robot. The target robot comprises a mechatronic mechanical arm with active joints kinematically corresponding to the exoskeleton modules and an actuator connected to these joints. The actuator of the target robot receives motion data from the exoskeleton encoders and controls the joints accordingly, providing replication of the user's movements during remote operation.

The disadvantages of the prototype are as follows:

- the exoskeleton lacks an additional degree of freedom to enable rotation of the user's torso, due to structural limitations and the configuration of elements in the back area;
- the exoskeleton lacks an additional degree of freedom to measure torso inclination of the user in the sagittal and frontal planes, also due to structural limitations and the configuration of elements in the back area;
- the degree of freedom provided for the pronation and supination angle of the hand relative to the forearm does not ensure the full range of angular rotation.

The technical problem addressed by the present invention, and its technical result, is the development of a Mechanical Motion-Capture Suit for Robotics, which enables the implementation of:

- an additional degree of freedom enabling measurement of the user's torso inclination relative to the vertical axis in the frontal plane (left-right);

– an additional degree of freedom enabling measurement of the user's torso inclination relative to the vertical axis in the sagittal plane (forward-backward);

– an additional degree of freedom enabling measurement of the user's torso rotational movement relative to the vertical axis;

– the capability to measure the full range of pronation and supination angles of the hand relative to the forearm.

The essence of the claimed technical solution is a Mechanical Motion-Capture Suit for Robotics, comprising a single-board computer designed to process data received from all encoders of the mechanical suit, as well as a power supply unit providing electrical power to all functional components of the device; the suit further includes a back fixture capable of mechanically attaching a third rotational joint with a built-in encoder and an adjustable-length third link configured to connect to the back area; the third link is adjustable in length, oriented vertically along the upper part of the back, and serves as the main supporting frame of the structure; a fourth link and a fifth link, both adjustable in length, are fixed horizontally to the third link in opposite directions, allowing rigid attachment and forming a frame for subsequent placement of the shoulder assemblies; at the ends of the fourth and fifth links, respectively, a fourth rotational joint with a built-in encoder and a fifth rotational joint with a built-in encoder are installed, configured to measure the arm abduction angle in the frontal plane; a sixth link and a seventh link, both curved and oriented towards the upper parts of the shoulder girdle, are each equipped at their ends with, respectively, a sixth rotational joint with a built-in encoder and a seventh rotational joint with a built-in encoder, configured to measure the forward arm elevation angle relative to the torso in the sagittal plane; the sixth and seventh rotational joints with built-in encoders are sequentially connected toward the elbow joint to an eighth rotational joint and a ninth rotational joint, each equipped with a built-in encoder and intended for measuring the rotational angle of the upper arm relative to its axis, **characterized in that** it additionally comprises a hip fixture configured to accommodate the single-board computer and the power supply unit; a first rotational joint with a built-in encoder mounted in the lower spine area, configured to measure lateral torso inclinations relative to the vertical axis; a second rotational joint with a built-in encoder mounted above the first rotational joint, configured to measure sagittal torso inclinations relative to the vertical axis; an adjustable-length first link positioned along the back and connected to a rotational shaft, which functionally connects the first and second links, both of which are adjustable in length; the shaft is configured to provide an additional axis of rotational movement; at the end of the second adjustable-length link, a third rotational joint with a built-in encoder is installed, providing measurement of torso rotational movement relative to the vertical axis; furthermore, adjustable-length eighth and ninth bearing-supported links are connected to the eighth and ninth rotational joints with built-in encoders and are fixed, respectively, to the first and second fixtures located on the lower third of the upper arms; the eighth and ninth bearing-supported links enable the transmission of rotational movement from the forearm to the eighth and ninth rotational joints with built-in encoders, allowing full anatomical mobility reproduction; at the ends of the adjustable-length eighth and ninth bearing-supported links, in the elbow area, a tenth and an eleventh rotational joint with built-in encoders are installed, configured to measure the flexion and extension angles of the arm at the elbow joint; in the direction of the forearm, tenth and eleventh adjustable-length links are connected, respectively, to the tenth and eleventh rotational joints with built-in encoders, enabling coordinated kinematic linkage and adjustment to the user's anatomical features; at the ends of the tenth and eleventh adjustable-length links, a twelfth rotational joint with a built-in encoder and a thirteenth rotational joint with a built-in encoder are installed, respectively, connected to a third wrist fixture and a fourth wrist fixture, and configured to measure the pronation and supination angles of the hand relative to the forearm; the twelfth and thirteenth links are fixed to the third and fourth wrist fixtures, respectively, and are equipped at their ends with a twelfth rotational joint with a built-in encoder and a thirteenth rotational joint with a built-in encoder, configured to measure the flexion-extension angles of the hand in the sagittal plane; the twelfth and thirteenth rotational joints with built-in encoders are further connected to a fourteenth link and a fifteenth link, respectively, which are fixed at their ends to a fifth and a sixth fixture mounted on the palms.

The claimed technical solution is illustrated by Fig. 1 to Fig. 3.

<u>Fig. 1</u> shows a schematic view of the back part of the mechanical suit.

<u>Fig. 2</u> shows a schematic overall view of the mechanical suit from the left side.

<u>Fig. 3</u> shows a schematic overall view of the mechanical suit from the right side.

<u>The positions in the figures indicate:</u>

1 – hip fixture;

2 – single-board computer;

3 – power supply unit;

4 – first rotational joint with built-in encoder;

5 – second rotational joint with built-in encoder;

6 – first adjustable-length link;

7 – rotational shaft;

8 – second adjustable-length link;

9 – third rotational joint with built-in encoder;

10 – back fixture;

11 – third adjustable-length link;

12 – fourth adjustable-length link;

13 – fifth adjustable-length link;

14 – fourth rotational joint with built-in encoder;

15 – fifth rotational joint with built-in encoder;

16 – sixth link;

17 – seventh link;

18 – sixth rotational joint with built-in encoder;

19 – seventh rotational joint with built-in encoder;

20 – eighth rotational joint with built-in encoder;

21 – ninth rotational joint with built-in encoder;

22 – eighth adjustable-length bearing-supported link;

23 – ninth adjustable-length bearing-supported link;

24 – first fixture on the lower third of the upper arm;

25 – second fixture on the lower third of the upper arm;

26 – tenth rotational joint with built-in encoder;

27 – eleventh rotational joint with built-in encoder;

28 – tenth adjustable-length link;

29 – eleventh adjustable-length link;

30 – twelfth rotational joint with built-in encoder;

31 – thirteenth rotational joint with built-in encoder;

32 – third wrist fixture;

33 – fourth wrist fixture;

34 – twelfth link;

35 – thirteenth link;

36 – twelfth rotational joint with built-in encoder;

37 – thirteenth rotational joint with built-in encoder;

38 – fourteenth link;

39 – fifteenth link;

40 – fifth palm fixture;

41 – sixth palm fixture.

Hereinafter, a description of the claimed technical solution is provided by the applicant.

The claimed Mechanical Motion-Capture Suit for Robotics is designed to be worn on the human body and is configured to measure and transmit the kinematic parameters of torso, shoulders, arms, elbows, forearms, and wrists movements. The suit comprises the following functional elements.

A hip fixture (1) is configured to accommodate a single-board computer (2), designed to process data received from all encoders of the mechanical suit, as well as a power supply unit (3), providing electrical power to all functional components of the device.

A first rotational joint with a built-in encoder (4) is mounted in the lower spine area and is configured to capture lateral (side-to-side) inclinations of the torso relative to the vertical axis.

A second rotational joint with a built-in encoder (5) is mounted above the first rotational joint (4) and is configured to measure sagittal inclinations of the torso (forward-backward) relative to the vertical axis.

A first adjustable-length link (6) is positioned along the back and connected to a rotational shaft (7).

The rotational shaft (7) functionally connects the first link (6) and the second link (8), both of which are adjustable in length, and provides an additional axis of rotational movement.

At the end of the second adjustable-length link (8), a third rotational joint (9) with a built-in encoder is installed, providing measurement of torso rotational movement relative to the vertical axis.

A back fixture (10) is designed for mechanical attachment of the third rotational joint (9) with a built-in encoder and the third adjustable-length link (11), thereby ensuring stable connection to the back area.

The third link (11), which is adjustable in length, is oriented vertically along the upper part of the back and serves as the main supporting frame of the structure. A fourth link (12) and a fifth link (13), both adjustable in length, are fixed horizontally to the third link (11) in opposite directions, providing rigid attachment and forming a frame for subsequent placement of the shoulder assemblies.

At the ends of the fourth link (12) and the fifth link (13), respectively, a fourth rotational joint with a built-in encoder (14) and a fifth rotational joint with a built-in encoder (15) are installed, configured to measure the arm abduction angle in the frontal plane.

A sixth link (16) and a seventh link (17) are curved and oriented toward the upper parts of the shoulder girdle. At the ends of these links, respectively, a sixth rotational joint with a built-in encoder (18) and a seventh rotational joint with a built-in encoder (19) are installed, configured to measure the forward elevation angle of the arms relative to the torso in the sagittal plane.

Sequentially connected to the sixth rotational joint with a built-in encoder (18) and the seventh rotational joint with a built-in encoder (19), in the direction of the elbow joint, are an eighth rotational joint (20) and a ninth rotational joint (21), each equipped with a built-in encoder and designed to measure the rotational angle of the upper arm relative to its axis.

Adjustable-length eighth bearing-supported link (22) and ninth bearing-supported link (23) are connected to the eighth rotational joint with a built-in encoder (20) and the ninth rotational joint with a built-in encoder (21), and are fixed, respectively, to the first fixture (24) and the second fixture (25), which are positioned on the lower third of the upper arm. The eighth bearing-supported link (22) and the ninth bearing-supported link (23) provide the transmission of rotational movement from the forearm to the eighth (20) and ninth (21) rotational joints with built-in encoders, thus achieving full reproduction of anatomical mobility.

At the ends of the adjustable-length eighth (22) and ninth (23) bearing-supported links, in the elbow area, a tenth rotational joint (26) and an eleventh rotational joint (27) with built-in encoders are installed, providing measurement of the flexion and extension angles of the arm at the elbow joint.

In the direction of the forearm, a tenth adjustable-length link (28) and an eleventh adjustable-length link (29) are connected, respectively, to the tenth rotational joint with a built-in encoder (26) and the eleventh rotational joint with a built-in encoder (27). These links are intended to provide coordinated kinematic linkage and adjustment to the user's anatomical features.

At the ends of the tenth (28) and eleventh (29) adjustable-length links, respectively, a twelfth rotational joint with a built-in encoder (30) and a thirteenth rotational joint with a built-in encoder (31) are installed, which are connected to the third wrist fixture (32) and the fourth wrist fixture (33), respectively. These joints are configured to measure the pronation and supination angles of the hand relative to the forearm.

A twelfth link (34) and a thirteenth link (35) are fixed to the third wrist fixture (32) and the fourth wrist fixture (33), respectively, mounted on the hands, and are equipped at their ends with a twelfth rotational joint with a built-in encoder (36) and a thirteenth rotational joint with a built-in encoder (37), configured to measure the flexion-extension angles of the hand in the sagittal plane.

A fourteenth link (38) and a fifteenth link (39) are connected, respectively, to the twelfth rotational joint with a built-in encoder (36) and the thirteenth rotational joint with a built-in encoder (37), and their ends are fixed to a fifth fixture (40) and a sixth fixture (41), mounted on the palms.

The elements of the claimed mechanical suit are structurally connected to each other, for example, by threaded connections (screwing), which ensures reliable fixation, the possibility of disassembly, and subsequent adjustment of individual units and components.

The following are examples of embodiments of the claimed technical solution.

Example 1. Use of the claimed Mechanical Motion-Capture Suit for Robotics.

In the initial position, the mechanical suit is worn by the user (human operator). For secure fixation and accurate reproduction of anatomical movements, all fixtures were fastened: the hip fixture (1), back fixture (10), shoulder fixtures (24 and 25) on the lower third of the upper arms, wrist fixtures (32 and 33), as well as palm fixtures (40 and 41), which are made of elastic fabric.

Each fixture was connected to the corresponding links and joints. The adjustable-length links (6, 8, 11, 12, 13, 16, 17, 28, 29, 34, and 35) provided precise length and geometry adjustment, eliminating play and ensuring full conformity to the body. All links were made of metal.

During lateral inclinations of the torso to the left and right, the first rotational joint (4) with a built-in absolute encoder registered the deviation relative to the vertical axis in the frontal plane, capturing the operator's lateral torso movements, thereby demonstrating the achievement of the claimed technical result.

During forward and backward inclinations of the torso, the second rotational joint (5) registered the deviation angle relative to the vertical axis in the sagittal plane. The rotational shaft (7) provided an additional degree of freedom to the links (6) and (8) and facilitated the operation of the mechanical suit during torso tilting movements, thereby demonstrating the achievement of the claimed technical result.

During torso rotation relative to the vertical axis, the third rotational joint (9) with a built-in absolute encoder recorded the angular displacement, providing measurement of the operator’s torso rotational movement, thereby demonstrating the achievement of the claimed technical result.

The fourth and fifth rotational joints (14) and (15) with built-in absolute encoders, located at the ends of the horizontal links (12) and (13), registered arm abduction and adduction in the frontal plane. The sixth and seventh rotational joints (18) and (19) with built-in absolute encoders, mounted on the curved links (16) and (17), measured forward arm elevation (sagittal lifting relative to the operator’s torso). The eighth and ninth rotational joints (20) and (21) with built-in absolute encoders recorded pronation and supination of the shoulder.

The tenth and eleventh rotational joints (26) and (27) with built-in absolute encoders, installed on the links (22) and (23) in the elbow area, recorded flexion and extension of the forearm relative to the upper arm.

The tenth and eleventh adjustable-length links (28) and (29), connected to the joints (26) and (27) with built-in absolute encoders, provided adjustment to the forearm length and optimal alignment of the links.

The twelfth and thirteenth rotational joints (30) and (31) with built-in absolute encoders, installed at the ends of the links (28) and (29) and connected to the wrist fixtures (32) and (33), recorded the full angle of pronation and supination of the hand relative to the forearm, thereby demonstrating the achievement of the claimed technical result.

The twelfth and thirteenth links (34) and (35) are additionally fixed to the wrist fixtures and equipped with joints (36) and (37) with built-in absolute encoders, which measured flexion and extension of the hand in the sagittal plane.

The fourteenth and fifteenth links (38) and (39), connected to the joints (36) and (37) with built-in absolute encoders and fixed to the palm fixtures (40) and (41), provided additional stabilization of the hand and transmitted forces to the palm segment, preventing undesirable parasitic displacements.

All angular data registered by the built-in absolute encoders are transmitted to the single-board computer (2) installed on the hip fixture (1), with power supplied to all encoders and the single-board computer from the power supply unit (3), also mounted on the hip fixture (1). The computer performed preliminary data processing, which was then transmitted via a wireless communication channel to the humanoid robot control system for synchronous motion reproduction (the "replay" mode). In a second variant, the data were stored in local memory for subsequent analysis and for training control algorithms of robotic systems (the "trajectory recording" mode).

Example 2. Real-time control of robot motion.

The mechanical suit is mounted on the operator's (human) body with fixation on the torso, shoulders, elbows, forearms, and hands. When the operator performs various movements (such as torso inclinations, arm elevation and abduction, forearm rotation, elbow and wrist flexion and extension), the built-in encoders register the corresponding angles in real time. The data are transmitted to the robot's control computer, which performs synchronous replication of the movements, ensuring precise following of the operator’s motions by the humanoid manipulator or the entire humanoid robot.

Example 3. Collection of kinematic data for training a robot control system.

The mechanical suit is fixed on the human body during the execution of a predefined set of movements (for example, lifting an object, performing household or industrial tasks). The angular values in all rotational joints are recorded to a file or transmitted to a cloud storage system. Subsequently, these data are used to create training datasets, to tune machine learning algorithms, or to adjust robot trajectories.

Example 4. Interactive control of a virtual avatar or animated character.

The data from the encoders are used to control a virtual model in a 3D software environment or in a VR/AR setting. This ensures high-precision motion replication, allowing the system to be used for character animation, motion capture preparation for games, films, or virtual presentations.

Thus, from the above description, it can be concluded that the identified **technical problem has been solved by the applicant and the claimed technical result has been achieved**, namely, the development of a device — a Mechanical Motion-Capture Suit for Robotics — which provides the following functional advantages:

– the presence of an additional degree of freedom enabling measurement of the user's torso inclination relative to the vertical axis in the frontal plane (left-right), which expands the functionality of the device when reproducing tilting movements;

– the presence of an additional degree of freedom enabling measurement of the user's torso inclination relative to the vertical axis in the sagittal plane (forward-backward), which expands the functionality of the device when reproducing tilting movements;

– the presence of an additional degree of freedom enabling measurement of the user's torso rotational movement relative to the vertical axis, which allows for the reproduction of torso twisting movements;

– the capability to measure the full angle of pronation and supination of the hand relative to the forearm, which enables anatomically accurate reproduction of hand movements around the forearm’s longitudinal axis.

Claims

A mechanical motion-capture suit for robotics, comprising:

a single-board computer configured to process data received from all encoders of the mechanical suit, as well as a power supply unit providing electrical power to all functional components of the device;

a back fixture configured for mechanical attachment of a third rotational joint with a built-in encoder and a third adjustable-length link configured to provide connection to the back area;

a third link, adjustable in length, oriented vertically along the upper part of the back, and serving as a supporting frame structure;

a fourth link and a fifth link, both adjustable in length, fixed horizontally to the third link in opposite directions, enabling rigid attachment and forming a frame for subsequent placement of the shoulder assemblies;

a fourth rotational joint with a built-in encoder and a fifth rotational joint with a built-in encoder are installed at the ends of the fourth link and the fifth link, respectively, configured to measure the arm abduction angle in the frontal plane;

a sixth link and a seventh link, both curved and oriented toward the upper parts of the shoulder girdle, with a sixth rotational joint with a built-in encoder and a seventh rotational joint with a built-in encoder installed at their respective ends, configured to measure the forward arm elevation angle relative to the torso in the sagittal plane;

an eighth rotational joint and a ninth rotational joint, each equipped with a built-in encoder and intended for measuring the rotational angle of the upper arm about its axis, are sequentially connected to the sixth and seventh rotational joints with built-in encoders in the direction of the elbow joint;

characterized in that it further comprises

a hip fixture configured to accommodate the single-board computer and the power supply unit;

a first rotational joint with a built-in encoder, mounted in the lower spine area, configured to capture lateral torso inclinations relative to the vertical axis;

a second rotational joint with a built-in encoder, mounted above the first rotational joint, configured to measure sagittal torso inclinations relative to the vertical axis;

a first adjustable-length link, positioned along the back and connected to a rotational shaft, which functionally connects the first and second links, both adjustable in length, the shaft being configured to provide an additional axis of rotational movement;

a third rotational joint with a built-in encoder, installed at the end of the second adjustable-length link, configured to measure torso rotational movement relative to the vertical axis;

wherein adjustable-length eighth bearing-supported link and ninth bearing-supported link are connected to the eighth and ninth rotational joints with built-in encoders and fixed, respectively, to a first fixture and a second fixture positioned on the lower third of the upper arm, the eighth and ninth bearing-supported links providing transmission of rotational movement from the forearm to the eighth and ninth rotational joints with built-in encoders, enabling full reproduction of anatomical mobility;

at the ends of the adjustable-length eighth and ninth bearing-supported links, in the elbow area, a tenth rotational joint and an eleventh rotational joint with built-in encoders are installed, configured to provide measurement of flexion and extension angles of the arm at the elbow joint;

in the direction of the forearm, a tenth adjustable-length link and an eleventh adjustable-length link are connected, respectively, to the tenth and eleventh rotational joints with built-in encoders, providing coordinated kinematic linkage and adjustment to the user's anatomical features;

at the ends of the tenth and eleventh adjustable-length links, a twelfth rotational joint with a built-in encoder and a thirteenth rotational joint with a built-in encoder are installed, respectively, connected to a third wrist fixture and a fourth wrist fixture, these joints being configured to measure the pronation and supination angles of the hand relative to the forearm;

a twelfth link and a thirteenth link are fixed to the third wrist fixture and the fourth wrist fixture, mounted on the hands, and are equipped at their ends with a twelfth rotational joint with a built-in encoder and a thirteenth rotational joint with a built-in encoder, configured to measure the flexion-extension angles of the hand in the sagittal plane;

a fourteenth link and a fifteenth link are connected, respectively, to the twelfth rotational joint with a built-in encoder and the thirteenth rotational joint with a built-in encoder, their ends being fixed to a fifth fixture and a sixth fixture mounted on the palms.

Abstract

Mechanical Motion-Capture Suit for Robotics

The invention relates to the field of robotics and mechatronics, specifically to wearable exoskeleton devices. The objective of the present invention is to create a kinematically identical device designed both for real-time control of humanoid robot movements and for collecting human motion kinematic parameters for their training.

Its essence is a mechanical motion-capture suit for robotics, comprising a single-board computer configured to process data received from all encoders of the mechanical suit, as well as a power supply unit providing electrical power to all functional components of the device; a back fixture is provided for mechanical attachment of a third rotational joint with a built-in encoder and a third adjustable-length link configured to connect to the back area; the third link is adjustable in length, oriented vertically along the upper part of the back, and serves as a supporting frame structure; a fourth link and a fifth link, both adjustable in length, are fixed horizontally to the third link in opposite directions, enabling rigid attachment and forming a frame for subsequent placement of the shoulder assemblies; at the ends of the fourth and fifth links, respectively, a fourth rotational joint with a built-in encoder and a fifth rotational joint with a built-in encoder are installed, configured to measure the arm abduction angle in the frontal plane; a sixth link and a seventh link, both curved and oriented toward the upper parts of the shoulder girdle, have at their respective ends a sixth rotational joint with a built-in encoder and a seventh rotational joint with a built-in encoder, configured to measure the forward arm elevation angle relative to the torso in the sagittal plane; an eighth rotational joint and a ninth rotational joint, each equipped with a built-in encoder and intended for measuring the rotational angle of the upper arm about its axis, are sequentially connected to the sixth and seventh rotational joints with built-in encoders in the direction of the elbow joint,

characterized in that it further comprises a hip fixture configured to accommodate the single-board computer and the power supply unit; a first rotational joint with a built-in encoder, mounted in the lower spine area, configured to capture lateral torso inclinations relative to the vertical axis; a second rotational joint with a built-in encoder, mounted above the first rotational joint, configured to measure sagittal torso inclinations relative to the vertical axis; a first adjustable-length link, positioned along the back and connected to a rotational shaft that functionally connects the first and second links, both adjustable in length, the shaft being configured to provide an additional axis of rotational movement; a third rotational joint with a built-in encoder, installed at the end of the second adjustable-length link, configured to measure torso rotational movement relative to the vertical axis; adjustable-length eighth and ninth

bearing-supported links connected to the eighth and ninth rotational joints with built-in encoders and fixed, respectively, to a first fixture and a second fixture positioned on the lower third of the upper arm, the eighth and ninth bearing-supported links providing transmission of rotational movement from the forearm to the eighth and ninth rotational joints with built-in encoders, enabling full anatomical mobility reproduction; at the ends of the adjustable-length eighth and ninth bearing-supported links, in the elbow area, a tenth rotational joint and an eleventh rotational joint with built-in encoders are installed, configured to measure the flexion and extension angles of the arm at the elbow joint; in the direction of the forearm, a tenth and an eleventh adjustable-length link are connected, respectively, to the tenth and eleventh rotational joints with built-in encoders, providing coordinated kinematic linkage and adjustment to the user's anatomical features; at the ends of the tenth and eleventh adjustable-length links, a twelfth rotational joint with a built-in encoder and a thirteenth rotational joint with a built-in encoder are installed, respectively, connected to a third wrist fixture and a fourth wrist fixture, these joints being configured to measure the pronation and supination angles of the hand relative to the forearm; a twelfth link and a thirteenth link are fixed to the third and fourth wrist fixtures mounted on the hands and are equipped at their ends with a twelfth rotational joint with a built-in encoder and a thirteenth rotational joint with a built-in encoder, configured to measure the flexion-extension angles of the hand in the sagittal plane; a fourteenth link and a fifteenth link are connected, respectively, to the twelfth and thirteenth rotational joints with built-in encoders, their ends being fixed to a fifth fixture and a sixth fixture mounted on the palms. 1 independent claim, 3 figures.

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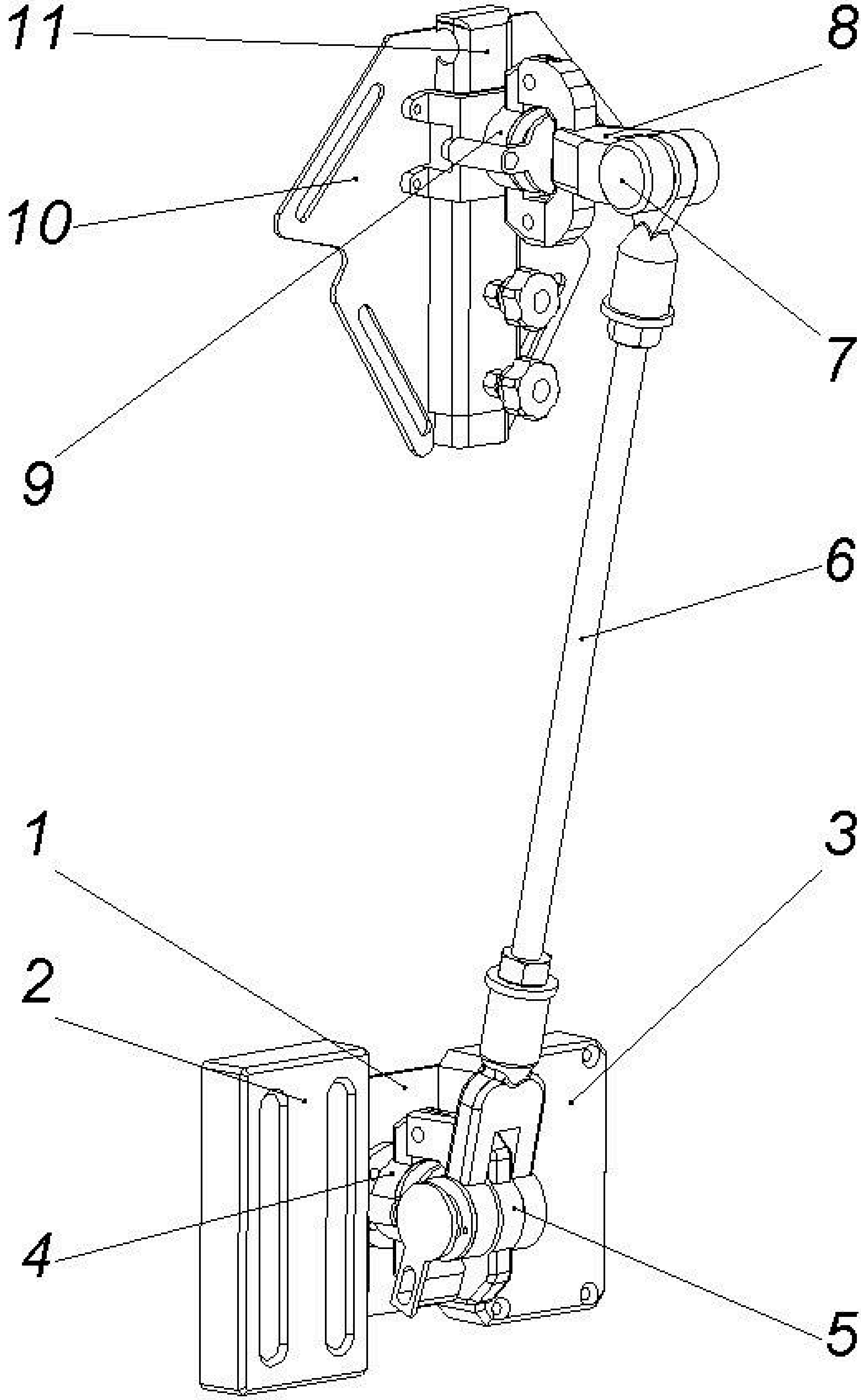

Fig. 1

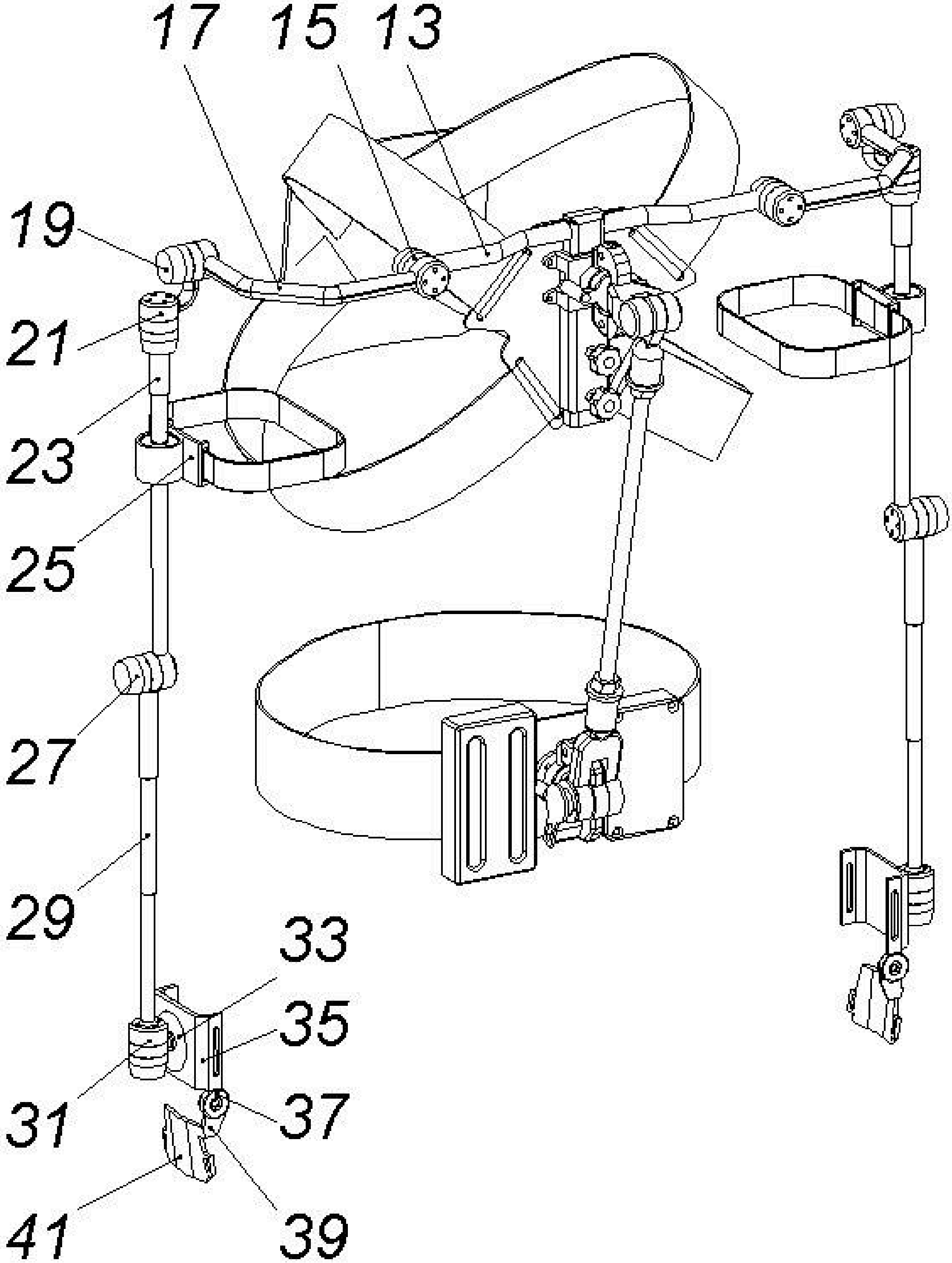

Fig. 2

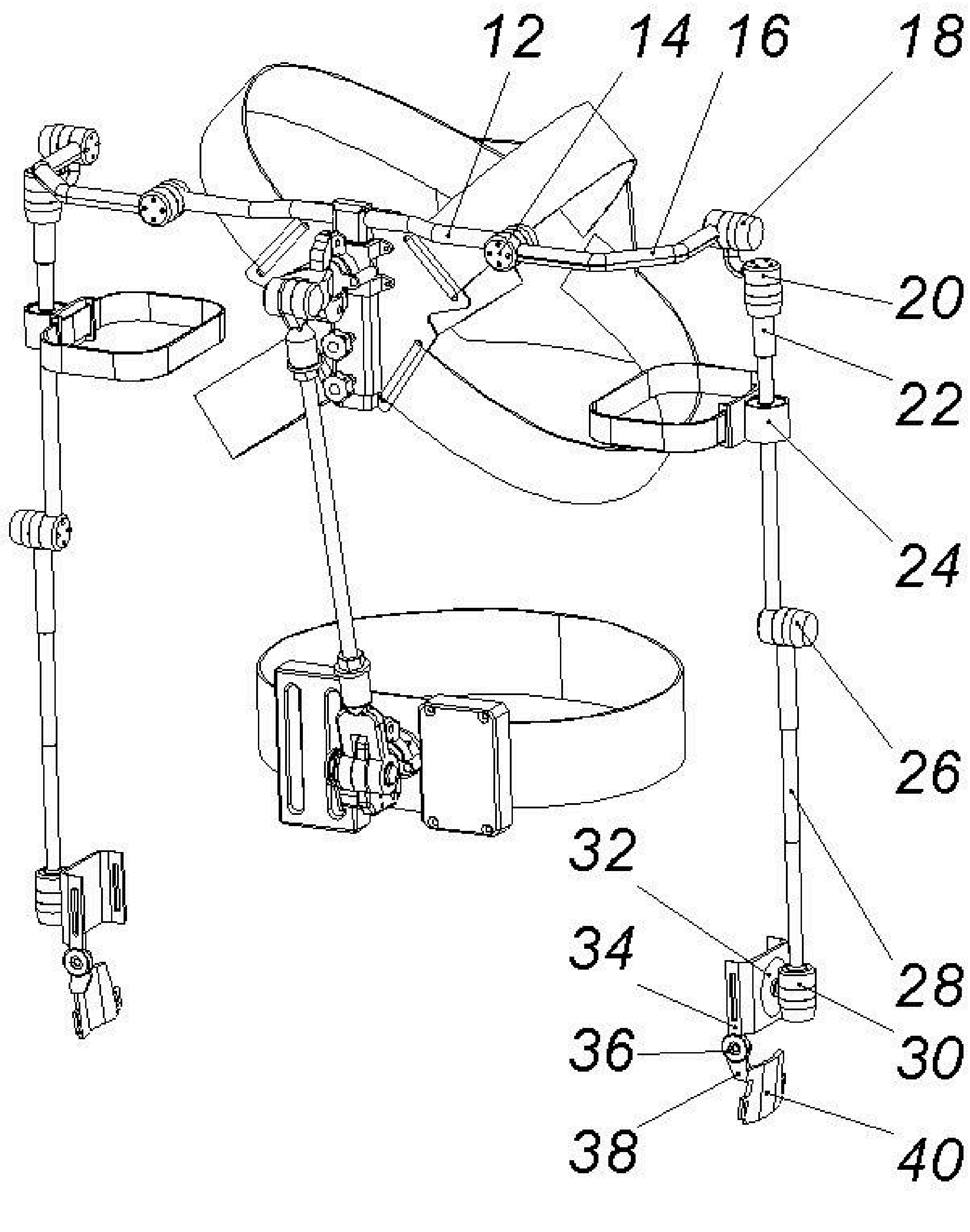

Fig. 3